Maurice Heide

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EDUCATION

ETH Zurich — Zurich, Switzerland
M.Sc. Data Science — Sep 2026 to present
- Coursework: Probabilistic AI, Mathematics of Data Science, Big Data, Seminar in Representation Learning

TU Dresden — Dresden, Germany
B.Sc. Computer Science, grade 1.0 (top grade), passed with distinction — Oct 2022 to Aug 2026
- Thesis (1.0) on radiology foundation models at Harvard Medical School and Massachusetts General Hospital
- All four mathematics exams 1.0 (30 ECTS); teaching assistant for Discrete Mathematics, best-rated group

Nanyang Technological University — Singapore
Exchange semester, GPA 4.83/5.00; the only exchange student of the relaunched partnership — Jan 2025 to May 2025

Seoul National University — Seoul, South Korea
Exchange semester, GPA 4.2/4.3 — Sep 2024 to Dec 2024
- One of two exchange places; A+ in Deep Learning (graduate level) and Statistics

EXPERIENCE

Harvard Medical School & Massachusetts General Hospital — Boston, USA
Visiting Undergraduate Researcher, Harvard University; Fintelmann Lab, Radiology — May 2026 to Sep 2026
- Bachelor thesis: benchmarked six published CT foundation models on 11 cohorts (20,862 scans) and distilled the two strongest into 14 compact 3D models (0.04M to 44M parameters)
- Built the training and evaluation stack: PyTorch DDP on up to 8 A100s with Slurm auto-resume (2,260 GPU-hours) and a tested evaluation package whose outputs match the original models byte for byte
- A 3M student is within 0.2 AUROC points of its 79M teacher (MLP probe), and the smallest runs 82.6× faster on CPU
- As the lab's only engineer, scaled its CT biomarker pipeline to 3,864 scans; checked its outputs against the images, caught and removed wrong values in 1,149 scans, and set the quality-control rules
- Took an LLM pipeline from design to production (Qwen3-4B on vLLM); it labels lung-cancer recurrence in clinical notes, with a review GUI for the medical team

Else Kröner Fresenius Center for Digital Health, TU Dresden — Dresden, Germany
Research Assistant, Kather Lab — Oct 2025 to Mar 2026, Nov 2023 to Jun 2024
- Shared first author of MuCoDi: distilled three pathology foundation models of up to 1.1B parameters into edge models; a 6.4M student, 99× smaller than the best teacher, is within 0.9 AUROC points of it on external data, and sub-million students run up to 605× faster on a Raspberry Pi 5
- Built a WebAssembly engine that predicts biomarkers from gigapixel slides in seconds, privately in the browser
- 2023 to 2024: turned 16 TB of whole-slide images into 23 MB of embeddings with a self-supervised pipeline

EY & Associés, Forensic & Integrity Services — Paris, France
Data Science Intern — Jul 2025 to Sep 2025
- Built a detector for AI-generated forgeries in EY's document-analysis platform (500+ clients, 25+ countries): compared 20 pretrained models, then fine-tuned DINOv3, using SVD to separate image content from forgery traces
- Curated a balanced training set of 250k real and 250k synthetic images from 6M, controlling for the JPEG and image-size biases that weaken detectors on unseen generators
- 99.8% accuracy on a public benchmark of 25 generators, 97 to 100% on passports and receipts forged by Nano Banana

Fraunhofer IVI — Dresden, Germany
Research Assistant, Applied Machine Learning — Mar 2023 to Sep 2023
- Built an automated calibration framework for the 29 near-infrared classifiers of a waste-sorting plant; XGBoost on 58 spectral parameters raised sorting accuracy by 22% for Germany's leading waste-management company
- Wrote an ETL pipeline whose statistical thresholds flag sensor drift and calibration errors, reducing plant downtime

PUBLICATION

T. Lenz[†], **M. Heide**[†], M. Gustav, N. G. Reitsam, J. N. Kather. *Multi-Teacher Contrastive Distillation for Edge-Efficient Pathology Foundation Models.* MICCAI 2026 Workshop on Efficient Medical AI. arXiv:2607.05533. [†]Equal contribution.

HONORS

- Best Paper Award, MICCAI 2026 Workshop on Efficient Medical Artificial Intelligence (EMA4MICCAI), Strasbourg
- Scholar of the German Academic Scholarship Foundation (Studienstiftung des deutschen Volkes) since 2022
- Hermann Medal for the best Abitur of the year (1.0); two prizes of the German Physical Society

SKILLS

Programming	Python (PyTorch, JAX, scikit-learn, XGBoost, pandas, Hugging Face, ONNX), SQL, Rust, C++, Java
Infrastructure	Slurm and Apptainer on GPU clusters, Docker, Kubernetes, GitHub Actions, pytest, mypy, W&B
Languages	German (native), English (C2, IELTS 8.5), French (A1)